



Alžběta Prášilová

Bioinformatician and Data Analyst with experience in biomedical research and an interest in computational modeling. Skilled in Python and interdisciplinary research, with contributions to peer-reviewed publications. Holds a Master's degree in Bioinformatics.



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prasilal



Alžběta Prášilová

Languages



Czech - native



English - C1



German - A2



Italian - A1

Programming skills

Languages: Python Julia R Bash

SQL* C#* Java*

Other skills: Git statistical analysis

Agent-based modeling HPC LaTeX

Machine learning*

*elementary knowledge

Interests



singing



climbing



playing musical instruments



running



hiking



animals



Dunajovická 112, Třeboň 37901,
Czech Republic



21/01/2000



Driving license: B



DofE Award: Silver level

Work experience

Independent Research Assistant (*Computational Immunology*)

2020 - now

[Mgr. Tomáš Bílý, MFF UK, Prague](#)

Worked on computational modeling projects and data analyses of immune system dynamics (Julia, Python). Contributed to poster presentations and participated in several international conferences (Presented a [poster](#) at [VPH 2024](#)).



Independent Data Analyst

Nov 2024 - now

Otorhinolaryngology research

[3rd Faculty of Medicine, Charles University, Prague](#)

Performed data analyses on patient data for projects in the field of Otorhinolaryngology. Co-authored a peer-reviewed paper on the [Characteristics of a healthy sinonasal microbiome \(2025\)](#).



Microbiology research

[Institute of Microbiology, The Czech Academy of Sciences](#)

Conducted experimental data preprocessing, statistical analyses, and visualization for experiments on algal nutrient response. Developed and evaluated simple machine learning models to predict biological outcomes based on environmental variables.



Neuromuscular research

[Motol University Hospital, Prague](#)

Processed patient data and performed exploratory/statistical analyses for a study on hereditary neuropathy.



Genomic Data Analyst

Jul 2021 - Sep 2021

[Mgr. Jan Janoušek, Ph.D., Centre Algatech, Třeboň](#)

Assisted in research focused on genomic data interpretation. Developed Python scripts to automate data analysis and support bioinformatics workflow



Education

Master's degree in Bioinformatics

2022 - 2025

Charles University, Prague

- ML, Genomics, Structural Bioinformatics, Mathematical Modeling
- Study Abroad: Erasmus at [Università di Torino](#), Turin (2023) 
- Master Thesis:

Under the supervision of [Yuriy Petrenko, PhD.](#), [Mgr. Tomáš Bílý](#) at the [Institute of Experimental Medicine of the CAS](#) and [BIOCEV](#)

I developed a 3D hybrid agent-based model of glioblastoma spheroid growth with integrated reaction-diffusion fields. Calibrated and qualitatively validated the system using GBM-MSD co-culture data to study how mesenchymal stromal cells influence tumor dynamics.



BIOCEV



Bachelor's degree in Bioinformatics

2019 - 2021

Charles University, Prague

- Statistics, Linear algebra, Programming (Python, C#, Java), Cell Biology, Genetics, UNIX, SQL
- Bachelor Thesis: [Tools for microbial community analysis using high-throughput amplicon sequencing data](#)



Other work & education experiences

Singer in a vocal group

2021 - 2025

[Petr Špaček & Band](#)

Performed across multiple Czech cities in annual holiday tours (audiences up to 1,000+). Gained experience in teamwork, stage presence, and public communication.

Event Hostess (Scientific conferences, Music Festivals)

2015 - 2018

Assisted with guest coordination, ticketing, and crowd guidance at large-scale events.

Exchange student (High school program)

2017 - 2018

[Lloyd C. Bird High School](#), Chesterfield, Virginia

High School: [Gymnázium Třeboň](#), Třeboň